Ebuka Philip Oguchi Ph.D.

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ACADEMIC APPOINTMENT

Assistant Professor (Tenure Track)
Department of Computer and Information Sciences, Spelman College
January 2026 – Present

RESEARCH INTERESTS

Cybersecurity and trust establishment for emerging wireless systems; message integrity and authentication; physical-layer security; location and RF fingerprinting; autonomous and vehicular networks; agricultural IoT; molecular and nano-scale communication security; applied machine learning for adversarial environments.

EDUCATION

- **University of Nebraska–Lincoln**
  Ph.D. in Computer Science and Engineering, 2025
  Advisor: Prof. Nirnimesh Ghose
  Dissertation: *Authentication and Message Integrity Verification for Emerging Wireless Networks*

- **Changchun University of Science and Technology, China**
  M.S. in Computer Applied Technology, 2020
  Advisor: Prof. Fang Ming
  Thesis: *Smoke Recognition Using ResNet and GoogleNet*

- **Nnamdi Azikiwe University, Awka, Nigeria**
  B.Eng. in Electronic and Computer Engineering, 2016
  Project: *Biometric Course Register Using Fingerprint Authentication*

PUBLICATIONS

Journal Articles (Under Review)

1. Oguchi, E.; Ghose, N.; Vuran, M.C. *Soil-Assisted Trust Establishment for Underground Internet-of-Things.* IEEE Transactions on Wireless Communications.
2. Oguchi, E.; Anderson, M.; Duong, T.T.; Wisniewska, A.; Ghose, N. *Systematization of Knowledge for Security in Molecular and Nano-Communications.* IEEE Transactions on Molecular, Biological, and Multi-Scale Communications.

Peer-Reviewed Conference Papers

1. Oguchi, E.; Ghose, N.; Vuran, M.C. *STUN: Secret-Free Trust Establishment for Underground Wireless Networks.* IEEE INFOCOM Workshops, 2022.
2. Oguchi, E.; Ghose, N.; Vuran, M.C. *VET: Autonomous Vehicular Credential Verification Using Trajectory and Motion Vectors.* EAI SecureComm, 2023.

Research Experience

Research Assistant, University of Nebraska–Lincoln 2021–2025

- Agricultural IoT Security: Secure bootstrapping and authentication for underground sensor networks using physical-layer features.
- Autonomous Vehicle Security: Message integrity verification using trajectory, motion vectors, and Doppler effects.
- RF Fingerprinting: Machine-learning-based authentication for underground and OTA environments.
- Molecular Communication Security: Lightweight security protocols for nano-scale biological communication.

Research Funding (As Part of Funded Projects)

- Research conducted as part of projects supported by the National Science Foundation (CNS-2331191, CNS-2225161, CNS-1619285, ECCS-2030272).
- Research conducted as part of projects supported by the Nebraska Center for Energy Sciences Research (NCESR).

Teaching Experience

- Teaching Assistant, Cryptography & Security (CSCE 477/877), University of Nebraska–Lincoln.
- Teaching Assistant, Computer Architecture and Software Systems, D.S. Adegbenro ICT Polytechnic.

Professional Service

- Reviewer: IEEE Transactions on Industrial Informatics; ACM Transactions on Cyber-Physical Systems.
- Reviewer: EAI SecureComm; ACM WiSec.
- Judge: UNL College of Engineering Poster Fair.

Mentorship

- Google CS Research Mentorship Program (CSRMP).
- Institute for African-American Mentoring in Computing Sciences (iAAMCS).
- Mentored Ph.D. and undergraduate researchers at UNL.

Awards

- Ph.D. Dissertation Award, University of Nebraska–Lincoln, 2025.
- Mary E. and Elmer H. Dohrmann Fellowship, 2023.
- Chinese Government Scholarship, 2018.

Skills

Machine learning (TensorFlow); Python, MATLAB, C; cryptographic protocol analysis; SDRs (USRP, GNU Radio); network security; incident response; Wireshark; and applied ML for adversarial systems.